

Kneser graphs $K(2k + n, k)$ with $n > 0$ are not CFON* 1-colorable:
the general proof

Shrey Shah

1 Efficient open dominating sets and CFON* 1-colorings

An efficient open dominating set of a Kneser graph K is a subset of the vertices, call it T , of the K graph where every vertex in K has exactly 1 neighbor in T .

Note that an efficient open dominating set is analogous to a CFON* 1-coloring because if T is the set of all colored vertices (since we only have one color, we can treat all colored vertices as the same).

Let each vertex, v , in K be assigned a variable:

$$x_v = \begin{cases} 1, & v \in T \\ 0, & v \notin T \end{cases}$$

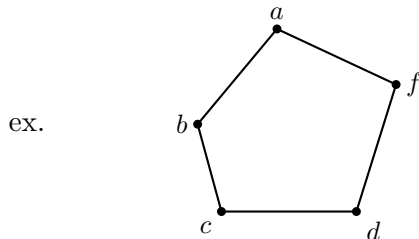
Thus, summing all the vertices in the open neighborhood $N(v)$ will equal 1.

2 The matrix equation

So for every vertex u in $N(v)$:

$$\sum_{u \in N(v)} x_u = \underbrace{0 + 0 + 0 + \dots}_{|N(v)|-1} + \underbrace{1}_{\text{the colored vertex}} = 1$$

Let A be a matrix with each row and column representing a vertex. If an element is in a row-column pair whose respective vertices are adjacent let it be equal 1, otherwise let it be zero.



$$\begin{matrix} & a & b & c & d & e & f \\ \begin{matrix} a \\ b \\ c \\ d \\ e \\ f \end{matrix} & \begin{bmatrix} 0 & 1 & 0 & 0 & 1 \\ 1 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 \\ 1 & 0 & 0 & 1 & 0 \end{bmatrix} \end{matrix}$$

Let x be a $|V(K)| \times 1$ matrix (vector) with each element representing a vertex, let the element equal 1 if the corresponding vertex is colored, and 0 otherwise.

$$Ax = A \begin{bmatrix} x_a \\ x_b \\ \vdots \\ x_f \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \longleftarrow \begin{array}{l} \text{number of colored vertices in the} \\ \text{neighborhood of the corresponding} \\ x\text{-vertex} \end{array}$$

The all-ones vector on the right is written $\mathbf{1}$ from here on.

3 The general case $K(2k + n, k)$

We generalize: $K(2k + n, k); k \geq 1, n > 0$.

The graph is $\binom{k+n}{k}$ -regular.

The adjacency matrix with each row and column corresponding to a vertex, with an element equalling 1 if the row and column's corresponding vertices are adjacent and 0 otherwise has $\binom{k+n}{k}$ 1's in each row. Let this matrix be M .

$$Mq = \mathbf{1}, \quad \text{suppose } q = \begin{bmatrix} g \\ g \\ \vdots \\ g \end{bmatrix}$$

Then,

$$\left(\frac{(k+n)!}{k!n!} \right) g = 1, \quad g = \frac{k!n!}{(k+n)!}$$

The eigenvalue

$$\lambda_j = (-1)^j \binom{k+n-j}{k-j}, \quad j = 0, 1, 2, \dots, k$$

So its magnitude is never 0 because $k+n-j \geq k-j \geq 0$ for every j , which makes every $\binom{k+n-j}{k-j}$ at least 1. Thus the eigenvalue for M is always non-zero, making M invertible.

Now suppose there was some vector w such that $Mw = \mathbf{1}$, and $w \in \mathbb{N}_0^{|V(K)|}$ such that it meets CFON* coloring restrictions.

Then:

$$\begin{aligned}
Mw &= Mq = \mathbf{1} \\
M^{-1}Mw &= M^{-1}Mq && \text{(we've proven invertibility)} \\
w &= q,
\end{aligned}$$

but q is fractional, because n is strictly greater than 0, which makes $\binom{k+n}{k} \geq k+1 \geq 2$ and so $g < 1$. This contradicts the definition of w . Thus, w cannot exist, and q is the unique solution.

Since q is the unique solution and all its elements are fractional, and since x_v can only equal 0 or 1 based on its inclusion in T , no efficient open dominating set exists and thus $K(2k+n, k)$ is not 1-colorable.